

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

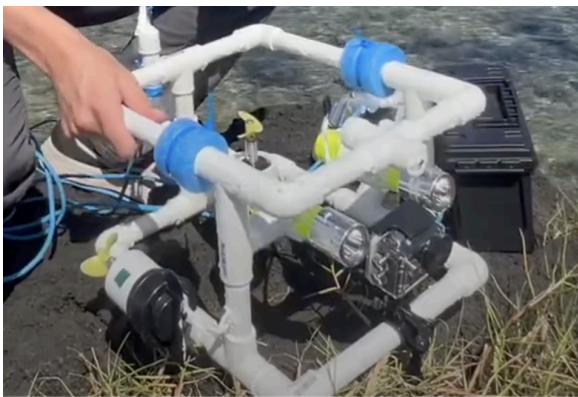
Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <ROV Inspiration Video #1>

 *Building a Remotely Operated Vehicle using PVC Pipe*

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some [free online journals and research databases](#)), professional organizations, government agencies, local and national newspapers, patents (here is a [patent search site for the United States Patent and Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).



Include at least one image of this prior design attempt. Make sure to cite all images.

Details: a design of a ROV the inspo was taken from

Include main details about this prior design attempt

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Analysis: We saw that this design of the frame was the most common among homemade ROVs, and decided that this video provided the best explanation and visual inspiration for us to create a somewhat close design of the frame.

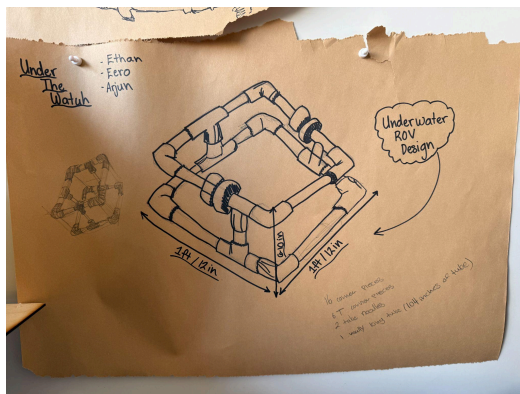
Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

At this point, we haven't done anything yet, but have taken inspiration for designs in the future.

Prior Design Attempt #2 - <ROV Design Sketch #2>

In Class Sketch

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some [free online journals and research databases](#)), professional organizations, government agencies, local and national newspapers, patents (here is a [patent search site for the United States Patent and Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).



Include at least one image of this prior design attempt. Make sure to cite all images.

We were inspired by the previous YouTube video and wanted to design support pillars along with pool floaties to add buoyancy.



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Include main details about this prior design attempt

Analysis: We drew the design to have the appropriate sizes where we can fit things inside of the ROV.

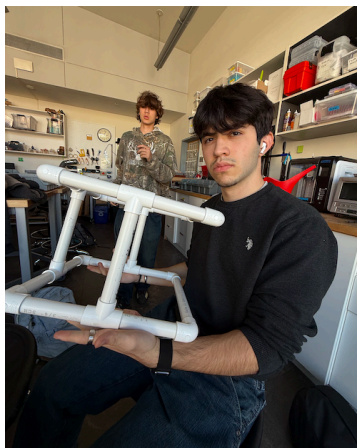
Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Figuring out the Correct sizing for everything, and what dimensions will be proportional to each other for a successful design of the frame. Dimensions for this frame were different from what is usually asked of for most common frames, ours was a bit off by an inch or so.

Prior Design Attempt #3 - <ROV Base Prototype #3>

In Class

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Include at least one image of this prior design attempt. Make sure to cite all images.

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Details: We chose to make 12 x 12 inch base with a height of 8 inches. We ensured there was 3 pillars to add additional strength to the frame. The weight could be difficult to work with but we will accommodate and work around that issue.

Include main details about this prior design attempt

Analysis: We created an ROV frame, this took a while through our cutting and measuring process. Our sizing is what we have planned to do from our previous sketch which formatted everything out.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

What was unsuccessful was that we did not include slot for the motors to be placed inside the frame. We prioritized the base so much that we skipped over creating a slot for the 3 motors we wanted to include. We had to redesign the frame to accommodate for the 3 motors.

Prior Design Attempt #4 - <insert name of prior design attempt #4>

Source

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some [free online journals and research databases](#)), professional organizations, government agencies, local and national newspapers, patents (here is a [patent search site for the United States Patent and Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

Analysis

Name: _____

Date: _____

Class: _____

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #5 - <insert name of prior design attempt #5>*Source*

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Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.